MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

Sl.No. 00066

SESSIONAL RECORD

DEPT.: COMPUTER SCIENCE ENGG

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SEMESTER: VII

SECTION: CSE-CORE-A

ROLL NO.: 225805222

SUBJECT: DEEP LEARNING

DATE OF TESTS: 14/10/2025

MAX MARKS : 15/20

MARKS OF: I TEST II TEST

MARKS FOR VALUATION - I TEST							MARKS FOR VALUATION - II TEST						
Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS	Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS
	a	b	c	d	e			a	b	c	d	e	
1							1						
							2						
							3						
							4						
							5						
6							6						
TOTAL IN FIGURES						30	TOTAL IN FIGURES						

IN WORDS ____________ IN WORDS ____________

RutBarbhai

Signature of the Candidate

Signature of the Teacher

Topic MCQs

QA1. option(c) ReLU

Q 2. option (b) It is prone to overfitting on large datasets

Q 3. option(b) To ensure that all features are scaled to a similar range.

Q 4. option(d) L2 Regularization

Q 5. option (c) Applying dropout regularization

Q 6. option (a) Decreasing the depth of the tree.

Q 7. option (b) implementing batch normalization

Q 8. option(a) To prevent overfitting by randomly disabling units during training

Q 9. option (c) Mean squared error (MSE)

Q 10. option (c) SMOTE (synthetic Minority over sampling Technique.

Answer-Key

Q1.	Q2.	Q3	Q4.	Q5	Q6	Q7	Q8.	Q9	Q10
C	B	B	D	C	A	B	A	C	C

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A.11 (a) A robust evaluation strategy for the housing price prediction model using ~~cv~~ cross validation would be as follows

1. choose performance Metrics :- For a regression task like predicting house prices, suitable metrics include Root Mean squared Error (RMSE), which penalizes large errors, Mean Absolute error (MAE), for a more direct interpretation of average error or R-squared (R^2) to see how much of price variance the model explains

2. Initial Data split :- First, split the entire dataset into a training set (eg 80%) and hold out test set (eg 20
The test set is set aside and only used for final evaluation of the model

3. K-Fold Cross Validation :- Apply K-fold cross validation on training data (a common choice is K=5 or K=10
. The training data is divided into K equal parts or "folds"

The model is trained K times. In each iteration, one fold is used for validation and remaining K-1 folds are used for training.

The choosen performance metrics is calculated on validation folds. after each iteration

4. **Aggregate Results**: After K iterations, ~~you~~ ~~you~~ you will have K performance scores. The final cross-validation performance is the average of these K-scores. This provides a more reliable estimate of model's generalization ability than a single train / validation split and is used to tune hyperparameter

5. **Final Evaluation**: once the best model configuration is found via cross-validation training set. Then evaluate its performance on the hold-out test set to get an unbiased estimate of how it will perform on new, unseen data.

A11 (b) Role of Activation Functions : The primary role of an activation function is to introduce non-linearity into a ~~linear~~ neural networks. A neural network layer performs ~~a no~~ a linear operation ($y = Wx + b$). Without a non-linear activation function, stacking multiple layers would be pointless, as a series of non-linear operations is mathematically equivalent to a single operation. By applying non-linear function, the network gains the ability to learn complex patterns and relationships in data enabling it to approximate any continious network.

Effect of No-Activation functions:

If a neural network had on activation functions between its layers the entire multi-layer network would be mathematically collapse into a single layer model. It would be capable of ~~linear~~ learning linear relationships, equivalent to performing linear regression. This would only be

~~be capable~~ limits its predictive power making it unable to solve complex problems that require modelling non-linear patterns, such as image recognition or natural language tasks

Q 12 A) Apply the backpropagation algorithm to a neural network...

A 12 A) The backpropagation algorithm trains a neural network by adjusting its weights based on error in its prediction the process involves two main passes for each batch of data:-

1. Forward Pass:- An input sample is fed into a network. It travels forward through the layers, with each layer calculating a weighted sum of its inputs and applying an activation function. This continues until the final output layer produces a prediction. The prediction value is then compared [illegible]

the actual target value to calculate the loss (error) using loss function like MSE.

2. Backward pass (Backpropagation):
This pass calculates how much each weight and bias in the network contributed to the final loss.
Gradient calculation: starting from the output layer, the algorithm uses the chain rule of calculus to compute the gradient of loss function with respect to the parameters of each layer, moving backward from the last layer to first. This gradient $\left(\frac{\partial L}{\partial w}\right)$ indicates the direction and the magnitude of the change needed for each weigh to reduce loss.

- Weight update:- once all the gradients are calculated, the network's weights and biases are updated using an optimization algorithm (like Gradient Descent). The update a weight w is

$$w_{new} = w_{old} - \eta\left(\frac{\partial L}{\partial w}\right)$$

Here η (eta) is the learning rate, a small hyperparameter that ~~learns~~ ~~learning~~ controls the step size of the update. This entire process is repeated for many epochs until the model converges

A12(b) Comparison of K-NN vs Decision ~~Tree~~ vs SVMs.

	Aspect	K-Nearest-Neighbours (K-NN)
1.	Learning Type	Lazy learner Memorizes data, no real training phase
2.	Decision Boundary	complex and non-linear
3.	Prediction Speed	Slow for large datasets (compares to all training pts)
4.	Interpretability	Low: prediction is based on neighbours not on explicit model
5.	Feature Scaling	It is a distance based algorithm

Factors For choosing K-NN

1. Small to Medium Data-set size: KNN's prediction time is slow, making it impractical for very large datasets

2. Low Dimensionality: It performs the best with a smaller number of features. Its performance degrades in high dimensional space due to "curse of dimensionality" where the concept of distance becomes less meaningful

3. Complex Decision Boundary: when the boundary seperating classes is highly irregular and cannot be easily approximated by a simple linear model or axis, parallel cut K-NNs flexibility is me advantage

4. No training time is priority: A "lazy learner" K-NN has a vir no training time. If you need deploy a model instantly and tolerate slower predictions KN Suitable

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